

Quantum Physics - Notes - v0.2.0

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Preface

Every theory section in these notes has been taken from the sources:

- Quantum Mechanics. [1]
- Fondamenti di fisica atomica e quantistica. [2]
- Modern Quantum Mechanics. [3]

About:

 [GitHub repository](#)

These notes are an unofficial resource and shouldn't replace the course material or any other book on quantum physics. It is not made for commercial purposes. I've made the following notes to help me improve my knowledge and maybe it can be helpful for everyone.

As I have highlighted, a student should choose the teacher's material or a book on the topic. These notes can only be a helpful material.

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1 From Wave-Particle Duality to Quantum States

1.1 Experimental Motivation: Wave-Particle Duality

The purpose of this section is not to give a history of the development of quantum mechanics, but rather to discover **what kind of mathematical theory we will eventually need** to describe the phenomena that we will discuss in these notes.

We begin from a very basic question: “*what is light?*”. Classically, there are two very different possibilities:

1. A **Particle** is something localized. At a given instant it **occupies a particular position, follows a trajectory, and arrives somewhere** as an individual object.
2. A **Wave**, on the other hand, is **spatially extended**. Different portion of the wave can overlap and interfere.

Historically, both descriptions were proposed for light. But for our purposes, the important question is not who proposed what. The important question is: **how can an experiment distinguish wave behaviour from particle behaviour?** In other words, definitions alone cannot choose between the two descriptions. So we **need an experimental criterion that allows us to distinguish between the two behaviors.**

🔗 What phenomenon is characteristic of waves? The wave vs. particle descriptions of light

Suppose we have light traveling from a source toward some obstacle and then onto a screen. **If light behaved only like classical particles,** the simplest picture would be:

source $\rightarrow$ particle trajectories $\rightarrow$ screen

Each particle follows some path, reaches a particular location, and contributes there.

But **waves behave differently** because different contributions can be added together. For two waves:

$$\psi_1, \quad \psi_2$$

The **combined disturbance**¹ is:

$$\psi_{\text{tot}} = \psi_1 + \psi_2$$

This simple operation has a crucial consequence: the **two waves may reinforce each other or cancel each other.** That phenomenon is called **Interference**.

¹The **total/resultant wave** produced when the two waves are present at the same place and time.

So already we have the first important distinction between waves and particles:

- **Particles** are localized objects.
- **Waves** are superposable² and extended disturbances.

At this point, “superposition” is still the familiar classical-wave idea.³ We are not yet stating the quantum superposition principle. Also, we have seen that the **interference phenomenon** is a characteristic of waves, but we have not yet seen any experimental evidence for it.

📖 Diffraction and interference as a signatures of waves

Interference is often observed experimentally through a **Diffraction** pattern. It is one of the **most convenient experimental manifestations of interference**. In general, interference is the general wave phenomenon, and diffraction is a **particular wave phenomenon in which a wave spreads after an aperture/obstacle, and the resulting diffraction pattern is produced by interference**. So we can have interference without diffraction, for example two independent water waves meeting each other.

🔍 Deepening: What does a “wave” actually have?

Before talking about light, think about something more intuitive: a **wave on a rope**. Imagine a rope that is fixed or held at the ends. If we move one point up and down, the disturbance propagates along the rope. The rope itself does **not** travel along with the wave. Each small piece of rope mainly moves up and down around its equilibrium position, while the **wave pattern** propagates horizontally.

Mathematically, at each point x along the rope, it has some displacement from equilibrium:

$$y(x, t)$$

Which is the description of the wave, where y is the vertical displacement of the rope at position x and time t ($y = 0$ is the equilibrium position). At one instant maybe $y = 2$ cm, and somewhere else $y = -1$ cm.

Thus, the quantity y is the **wave amplitude** at that point and instant. It indicates how far the rope is displaced from its equilibrium position.

For the case of **light**, instead of writing $y(x, t)$, we can think about:

$$E(x, t)$$

So the “amplitude” is the **value of the oscillating wave quantity**.

²They can be added together. More precisely, in a linear wave system, if ψ_1 and ψ_2 are allowed waves, then $\psi_1 + \psi_2$ is also an allowed wave.

³The sum of two waves is still a wave.

Deepening: What happens when two waves meet?

This is the defining ingredient behind interference. Suppose two waves arrive at the same place. Their amplitudes add:

$$E = E_1 + E_2$$

This is the superposition principle for classical waves. Now imagine at instant:

$$E_1 = +1, \quad E_2 = +1$$

Then:

$$E = +2$$

They reinforce each other. But perhaps instead:

$$E_1 = +1, \quad E_2 = -1$$

Then:

$$E = 0$$

They cancel each other. That is interference. So the fundamental thing that interference is **the amplitude**: $E_1 + E_2$.

Suppose light hits a screen. We don't see the electric field E directly:

$$E = +3, \quad E = -2, \quad E = +1, \quad \dots$$

Because the electromagnetic field oscillates extremely rapidly, and our eyes cannot follow it. Instead, the screen receives **energy** from the light.

- A region receiving more energy appears brighter.
- A region receiving less energy appears darker.

The quantity describing this is called the **Intensity** I , the **energy delivered per unit area per unit time**, roughly speaking.

For a light wave, intensity is proportional to $\propto$ the square of the wave amplitude:

$$I \propto |E|^2$$

❓ Why square the amplitude? One immediate reason the square makes sense is that amplitude can have a sign (e.g., $E = +1$ or $E = -1$), but **energy cannot be negative**. Squaring guarantees that the intensity is always positive. The sign is important for **interference**, but the final measured energy/intensity is always positive. Thus:

1. Add the amplitudes of the waves to get the total amplitude.
2. Square the total amplitude to get the intensity.

 Example 1: Interference of two waves

Imagine each wave alone would have amplitude magnitude 1. If only wave 1 existed:

$$E_1 = 1$$

So its intensity would be proportional to:

$$I_1 \propto 1^2 = 1$$

Similarly, if only wave 2 existed:

$$E_2 = 1$$

So its intensity would be proportional to:

$$I_2 \propto 1^2 = 1$$

We might naïvely think that two waves together should therefore give:

$$I = I_1 + I_2 = 1 + 1 = 2$$

But that is **not necessarily true**. We must first add their amplitudes.

- **Case A: the waves reinforce each other.** If:

$$E_1 = 1, \quad E_2 = 1$$

Then:

$$E = E_1 + E_2 = 1 + 1 = 2$$

So the intensity is:

$$I \propto |E|^2 = 2^2 = 4$$

We have **constructive interference**.

And if we had:

$$E_1 = -1, \quad E_2 = -1$$

Then:

$$E = E_1 + E_2 = -1 - 1 = -2$$

So the intensity is:

$$I \propto |E|^2 = |-2|^2 = 2^2 = 4$$

Again, we have **constructive interference**, even though the amplitudes are negative. The sign of the amplitude does not matter for the intensity.

- **Case B: the waves cancel each other.** If:

$$E_1 = 1, \quad E_2 = -1$$

Then:

$$E = E_1 + E_2 = 1 - 1 = 0$$

So the intensity is:

$$I \propto |E|^2 = 0^2 = 0$$

So the screen is dark there. This is **destructive interference**.

Let's generalize the previous example. Imagine one sinusoidal wave reaching some point P :

$$E_1(t) = E_0 \cos(\omega t)$$

Its average intensity is proportional to the square of the amplitude:

$$I_1 \propto E_0^2$$

So far, nothing strange. Now suppose **two waves** reach the same point P . Because ordinary wave equations are linear, the physical field is the sum:

$$E(t) = E_1(t) + E_2(t)$$

This is the classical **superposition principle for waves**. But intensity is not proportional to E , but rather to the **square** of the resulting amplitude:

$$I \propto |E|^2 \implies I \propto |E_1 + E_2|^2$$

Expanding:

$$|E_1 + E_2|^2 = |E_1|^2 + |E_2|^2 + 2\Re(E_1^* E_2)$$

The last term $2\Re(E_1^* E_2)$ is called the **Inference Term**. This term is the reason why:

$$|\psi_1 + \psi_2|^2 \neq |\psi_1|^2 + |\psi_2|^2$$

Or in words, the **intensity of two waves together is not equal to the sum of their individual intensities**. The inference term can be:

- **Constructive Interference** if the phases agree:

$$2\Re(E_1^* E_2) > 0$$

The two contributions reinforce each other. We have a **maximum** in the intensity.

- **Destructive Interference** if the phases disagree:

$$2\Re(E_1^* E_2) < 0$$

The contributions oppose each other and may cancel. We have a **minimum** in the intensity.

So an **alternating pattern of maxima and minima is a signature of interference**, and it is a **signature of wave-like behavior**. Therefore, observing such a diffraction/interference pattern provides an experimental criterion for identifying the wave-like nature of light.

Deepening: The interference term

Let's remove for a moment the absolute value from each term. Then the intensity is still proportional to the square of the amplitude:

$$I \propto (E_1 + E_2)^2$$

Expanding it exactly like ordinary algebra:

$$(E_1 + E_2)^2 = E_1^2 + E_2^2 + 2E_1E_2$$

The last term $2E_1E_2$ is the **Interference Term** in the simplest real-valued case. It is really **important** because without it, we would just have:

$$E_1^2 + E_2^2$$

Which is just “intensity from wave 1 plus intensity from wave 2”. The extra term tells us what happens when the two waves meet.

For example, if $E_1 = 1$ and $E_2 = 1$, then the intensity is:

$$I \propto (1 + 1)^2 = 4$$

But separately, the intensity from each wave is:

$$I_1 + I_2 = 1^2 + 1^2 = 2$$

The extra term $+2$ comes from the interference term:

$$2E_1E_2 = 2 \cdot 1 \cdot 1 = 2$$

So the waves reinforce each other. Now take $E_1 = 1$ and $E_2 = -1$. Then the intensity is:

$$I \propto (1 - 1)^2 = 0$$

But separately, the intensity from each wave is:

$$I_1 + I_2 = 1^2 + (-1)^2 = 1 + 1 = 2$$

The extra term -2 comes from the interference term:

$$2E_1E_2 = 2 \cdot 1 \cdot (-1) = -2$$

So the interference term is simply what can either increase the intensity or decrease it.

🔍 Now, why did we previously write the interference term as $2\Re(E_1^*E_2)$ instead of $2E_1E_2$? Because for light waves we often represent the amplitude using complex numbers. Then the correct version of the same algebra becomes:

$$\begin{aligned} |E|^2 &= |E_1 + E_2|^2 \\ \downarrow |z|^2 &= z \cdot z^* = (x + iy)(x - iy) = x^2 + y^2 \\ \downarrow \text{if we assume } E_1 + E_2 &= z \\ &= (E_1 + E_2)(E_1 + E_2)^* \\ &= (E_1 + E_2)(E_1^* + E_2^*) \\ &= E_1E_1^* + E_1E_2^* + E_2E_1^* + E_2E_2^* \\ \downarrow E_1E_1^* &= |E_1|^2, \quad E_2E_2^* = |E_2|^2 \\ &= |E_1|^2 + |E_2|^2 + E_1E_2^* + E_2E_1^* \end{aligned}$$

Now the last two terms are complex conjugates of each other. Let:

$$z = E_1 E_2^*$$

Thus its complex conjugate is:

$$z^* = (E_1 E_2^*)^* = E_1^* (E_2^*)^* = E_1^* E_2$$

Therefore:

$$E_1 E_2^* + E_2 E_1^* = z + z^*$$

And for any complex number:

$$z = a + ib$$

We have:

$$z^* = a - ib$$

So:

$$z + z^* = a + ib + a - ib = 2a$$

But a is the real part of z , represented as $\Re(z)$. So we have:

$$z + z^* = 2\Re(z)$$

Therefore:

$$E_1 E_2^* + E_2 E_1^* = 2\Re(E_1 E_2^*) \equiv 2\Re(E_1^* E_2)$$

And conceptually it is just the same cross term as $2E_1 E_2$.

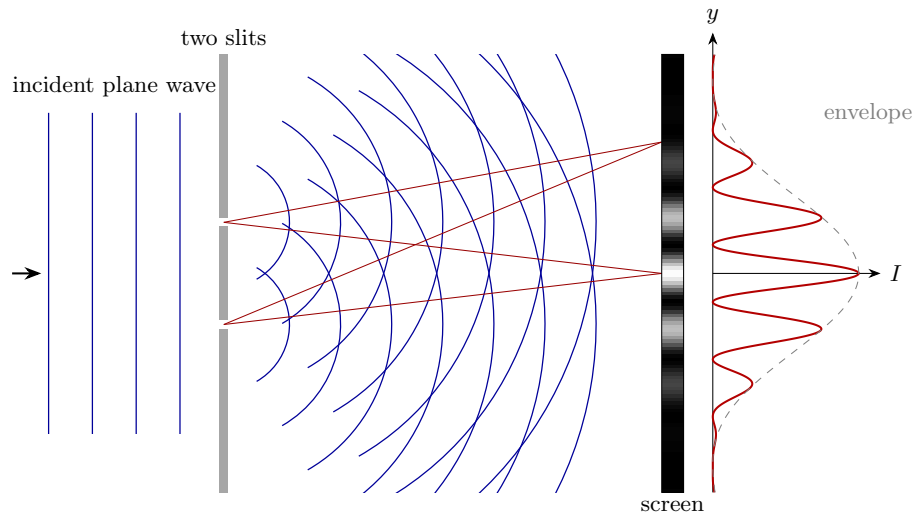


Figure 1: **Diffraction and interference in a two-slit experiment.** An incident plane wave passes through two slits; the emerging waves spread, overlap, and interfere, producing an intensity pattern on the screen modulated by a diffraction envelope.

Deepening

Some details about the Figure 1:

- ❓ **What is the “incident plane wave”?** On the left we have several **parallel blue vertical lines**, those lines are **wavefronts of the same incoming wave**.

A wavefront is a set of points where the wave is in the same state of oscillation. We do not need to know particular details; visually, think of each blue line as one crest/front of the wave. Because these wavefronts are straight and parallel, we call it a **plane wave**. So “*incident plane wave*” simply means **a wave arriving at the barrier with flat, parallel wavefronts**. “*Incident*” just means **incoming**.

Finally, the arrow indicates that the wave propagates from left to right.

- ❓ **What are the “two slits”?** The grey vertical object is a barrier with **two small openings**. Most of the wave is blocked by the barrier, only the portions reaching those openings pass through.

Here, the crucial observation is what happens **after** each slit. Instead of continuing only straight ahead, the wave **spreads out**. That’s why the figure draws curved blue wavefronts coming from each slit.

This spreading is the **diffraction** part! Each slit effectively gives us a new spreading wave. And now we have precisely what we need for **interference**: two waves reaching the same points on the screen (black rectangle on the right).

- ❓ **What happens between the slits and the screen?** Pick some point P on the screen. There are two wave contributions reaching P : ψ_1 from the upper slit and ψ_2 from the lower slit. The red straight lines in the figure illustrate examples of those two paths.

The two contributions add:

$$\psi(P) = \psi_1(P) + \psi_2(P)$$

But depending on the point P , the two paths have different lengths. So the waves may arrive “*in phase*” (reinforce, constructive interference), “*out of phase*” (cancel out, destructive interference), or somewhere in between (weaker reinforcement or weaker cancellation). And that’s why different points of the screen have different brightness.

- ❓ **Why is the center of the screen bright?** Call the two slit-to-screen distances r_1 and r_2 . At the exact center of the screen, by symmetry, the two distances are equal:

$$r_1 = r_2$$

Therefore the two waves have traveled the same distance, and they reinforce each other:

$$\psi_1 + \psi_2 \implies I \propto |\psi_1 + \psi_2|^2$$

So the intensity is large and the **central bright band corresponds to constructive interference.**

- ❓ **What are the dark bands?** Move upward or downward along the screen. Now one slit is slightly closer than the other:

$$r_1 \neq r_2$$

At certain position, the difference in traveled distance causes one wave to arrive exactly opposite in phase to the other.

For example:

$$\psi_1 = A, \quad \psi_2 = -A$$

Therefore:

$$\psi_1 + \psi_2 = A - A = 0$$

And hence the intensity is zero:

$$I \propto |\psi_1 + \psi_2|^2 = 0$$

So that location is dark, corresponding to **destructive interference.**

- ❓ **What is that red curve on the right?** The horizontal direction of the red graph represents the **intensity** I measured on the screen, while vertically we move along the screen coordinate y .

So where the solid red curve extends far to the right, the **intensity is large and the screen is bright.** Where it comes **close to zero** $I \approx 0$, **the screen is dark.**

So the oscillating solid red curve is essentially a **graph of $I(y)$, the intensity as a function of vertical position on the screen.**

- ❓ **What is the “envelope”?** Look at the **dashed red curve** labelled “envelope”. It is **not** “*the level of interference*”. Instead, it **describes how large the interference peaks are allowed to become at different positions.**

A more natural question is: *why are the bright fringes near the center stronger than the bright fringes farther away?* What is exactly what the dashed curve in figure is trying to show. The red solid curve $I(y)$ oscillates between bright and dark because of the **interference between the two slits.**

But notice also that the heights of the bright peaks are not all the same. Near the center, the peaks are large; farther away, they become smaller. The dashed curve simply joins the tops of those possible peaks. So the **envelope** is the **smooth curve giving the**

maximum possible intensity at each position on the screen (at each y). In other words, it is just a boundary for the interference pattern.

❓ **But why does the envelope exist?** Because the slits are not mathematical points; they have a finite width. Even **one slit by itself** produces a diffraction (i.e., spreading) pattern on the screen. So each slit does not send the same wave strength equally in every direction. It sends most strongly approximately forward, around the center of the screen, and more weakly at larger angles.

Therefore, before we even combine the two slits, each slit already has something like: strong near the center, weaker farther away. That broad distribution is the **single-slit diffraction pattern**. Now, if we combine the two slits, the two effects happen simultaneously and each slit diffracts; then the waves from the two slits interfere and the final pattern is the product of the two effects:

- **Single-slit diffraction** gives the broad envelope.
- **Two-slit interference** gives the oscillating pattern within that envelope.

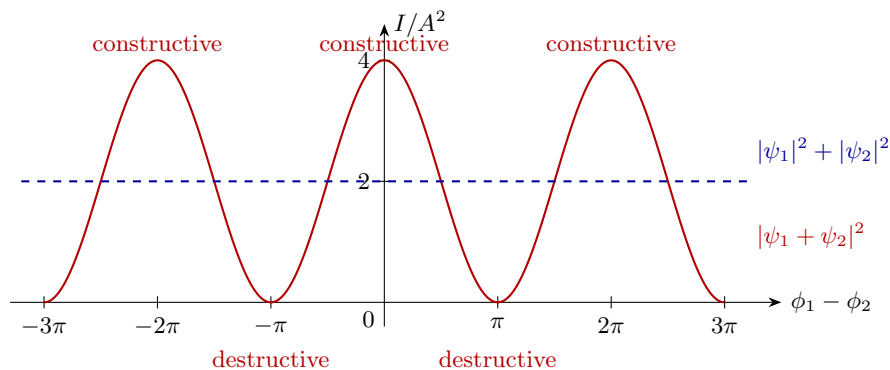


Figure 2: **The interference term.** The resulting intensity depends on the relative phase $\phi_1 - \phi_2$: constructive interference produces maxima, while destructive interference produces minima.

⚙️ Electron diffraction

Now we take the criterion we just established for waves (i.e., the presence of interference)

diffraction/interference $\implies$ wave-like behaviour

And apply it to something that, classically, we would expect to behave as a particle: **the electron**.

The surprising experimental observation is that a beam of electrons, when sent onto a crystalline sample, produces a **pattern that looks essentially like the diffraction pattern produced by light**.

💡 **The idea.** Before this experiment, the classical expectation was: electron behaves $\approx$ particle. *Why?* Because electrons have properties we normally associate with particles, they have mass, charge, momentum, and can form an electric current.

But now we perform an experiment designed to test for wave behaviour. We send electrons toward an ordered crystalline structure. And experimentally we observe, again, a **diffraction pattern**. And from the previous discussion, we know that it means:

diffraction pattern $\implies$ interference $\implies$ wave-like behaviour

So the immediate conclusion is that **electrons behave like waves**.

❓ **Why is this surprising?** Because the same physical object (the electron) was already well described as a particle in other experiments. So we have:

$$\text{electron} \quad \begin{cases} \text{particle-like in some experiments} \\ \text{wave-like in diffraction experiments} \end{cases}$$

This is already the beginning of **wave-particle duality**. However, we shouldn't jump ahead just yet. We'll see it in more detail later. At this point, all we have established is that **an electron beam produces a wave-like diffraction pattern**.

❓ What exactly do we observe?

1. The electron source produces a collimated beam of electrons.
2. These electrons hit a crystal.
3. A crystal contains atoms arranged in a regular periodic structure.
4. The electrons interact with that periodic structure, and the detected electrons are not distributed uniformly everywhere. Instead, they **appear preferentially in specific directions**, giving a pattern of bright spots.

That is exactly the kind of spatial organization we expect from diffraction. The regular atomic arrangement plays, for electrons, a role analogous to a diffraction grating for light.

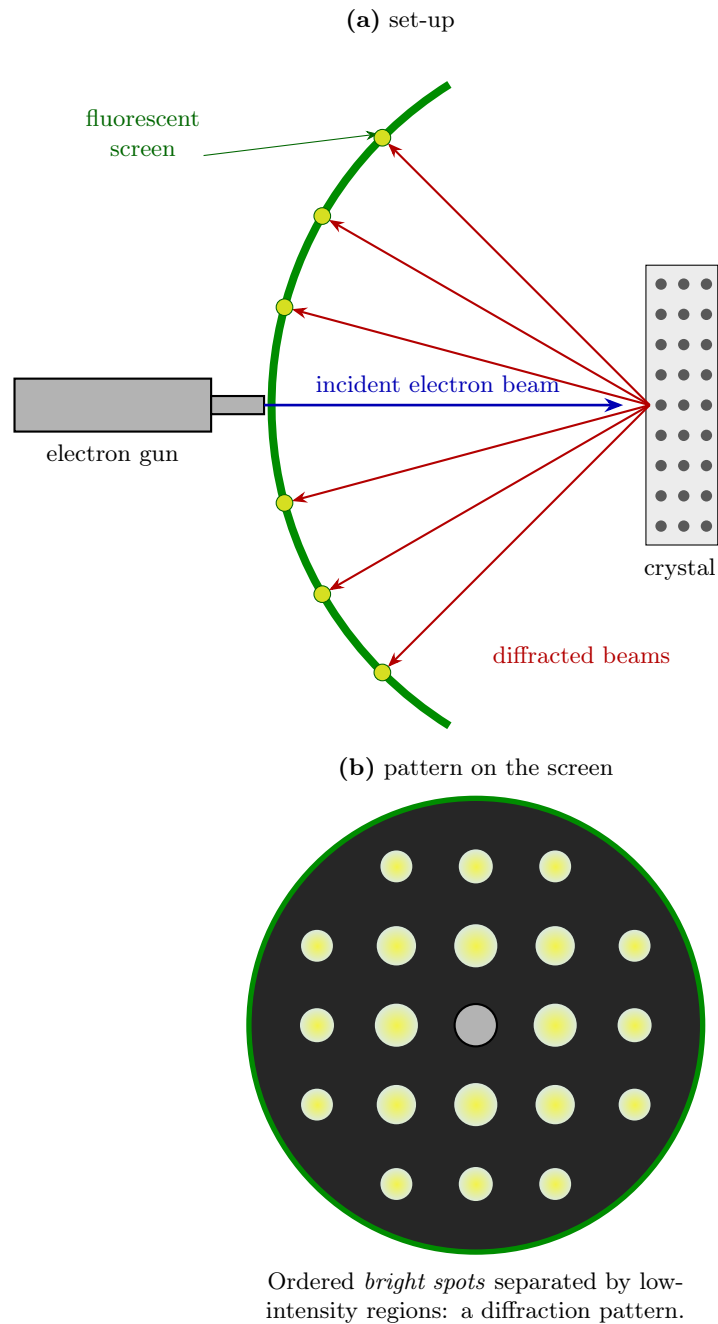


Figure 3: **Electron diffraction from a crystal.** A collimated electron beam produced by an electron gun is incident on a crystalline sample. The scattered electrons are detected on a fluorescent screen, where they form an ordered diffraction pattern with bright maxima separated by low-intensity regions.

In Figure 3, we show a schematic of the experimental set-up and the resulting diffraction pattern.

- In panel (a), the **electron gun** is the **source**. It emits a collimated beam of electrons **toward the crystal**. The crystal is not just a random obstacle: its atoms are arranged periodically, so it behaves like a diffraction structure for the electron beam.

The **arrows** leaving the crystal represent the **diffracted directions**. They are not meant to say that we know a classical trajectory for each electron; they indicate the **directions where the scattered electron signal is enhanced because of diffraction**.

Those **electrons then hit the fluorescent screen**. The screen converts the electron impacts into visible light: the electrons excite the fluorescent material, which then emits light.

- Panel (b) shows what we actually observe on that screen: a **set of bright spots at specific positions**. Those bright spots are the diffraction maxima. Between them, the signal is much weaker or absent. That ordered pattern is the important experimental result.

References

- [1] A. Berera and L.D. Debbio. *Quantum Mechanics*. Cambridge University Press, 2021.
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